Christian Scaff

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Education

Columbia University — Sept 2025 – Dec 2026
M.S. in Computer Science (expected Dec 2026) — GPA: 3.62/4.0

University of Florida — Aug 2021 – May 2025
B.S. in Computer Science — GPA: 3.96/4.0

Publications

PALS: Preference-guided Active Automata Learning for Symbolic Reinforcement Learning in Games
Williams Fishell, **Christian Scaff**, Sam Kouteili, and Mark Santolucito. *NExT-Game: New Frontiers in Game-Theoretic Learning*, ICML 2026 Workshop
openreview.net/forum?id=jRUJpASPQJ

Mining Beyond the Bools: Learning Data Transformations and Temporal Specifications
Sam Kouteili, Williams Fishell, **Christian Scaff**, Mark Santolucito, and Ruzica Piskac. To appear, *ESOP 2027*
arxiv.org/abs/2603.06710

Experience

Graduate Research Assistant — Sept 2025 – Present
Columbia University – Barnard PL Laboratory, advised by Prof. Mark Santolucito — *New York, NY*

- Led a team of three students, developing a cloud FPGA platform for agentic hardware design: seven Lattice ECP5-85F boards running LiteX/VexRiscv SoCs, a FastAPI and Redis orchestrator, and on-demand AWS Fargate build tasks.
- Published *manhattan-reasoning-gym* to PyPI, a Python SDK and CLI for containerized Yosys and nextpnr synthesis, sandboxed agent execution, and remote FPGA programming.
- Developing formally verified benchmarks and reinforcement learning environments measuring if LLM agents can optimize RTL for PPA (Performance, Power, Area).

Entrepreneurial Lead — Jun 2026 – Aug 2026
NSF I-Corps – Manhattan Reasoning — *New York, NY*

- Led customer discovery for a venture commercializing formally verified reinforcement learning environments for RTL design tasks.
- Conducted 100 interviews with hardware engineers, RL vendors, AI researchers, and enterprise customers to assess the market demand.

Research Intern — May 2025 – Aug 2025
Georgia Tech Research Institute — *Atlanta, GA*

- Developed Wy-Graph, a Python tool that translates LLVM IR programs into knowledge graphs, conveying program semantics and syntax through 28 node types and 41 edge types.
- Designed the graph schema to produce training data for neuro-symbolic program analysis, targeting differentiable inductive logic programming over program structure.

Research Programmer — Aug 2024 – May 2025
University of Florida GATAS Laboratory — *Gainesville, FL*

- Developed four domain-specific languages (DSLs) for AlgebraicJulia, an open-source Julia software ecosystem for applied category theory.
- Converted three handwritten DSLs for ACSets.jl, Catlab.jl, and DiagrammaticEquations.jl into parsing expression grammars (PEGs), improving their maintainability and accessibility.
- Implemented a DSL for cellular sheaf creation in AlgebraicOptimization.jl with MLStyle.jl, enabling the definition of cellular sheaf objects in a more intuitive and mathematical manner.

Technology Architecture Analyst — Jun 2024 – Aug 2024
Accenture — *Chicago, IL*

- Automated IT monthly reporting for a healthcare client via Power BI, reducing hours of manual analysis.
- Analyzed five months of CMDB data from ServiceNow and consolidated insights into an interactive Power BI dashboard.
- Monitored incident trends, change requests, and SLA metrics across onshore/offshore teams.

Independent Research

Equivalence Checking LLM-Optimized GPU Code Spring 2026
Columbia University – advised by Prof. Stephen A. Edwards *New York, NY*

- Investigated whether LLM-generated CUDA optimizations can be proven to preserve the semantics of the C++ programs they replace.
- Evaluated CBMC and LLM-driven Lean 4 proofs as candidate approaches for the optimization front-end of MACCHIATO, a DARPA MOCHA compiler toolchain.

Teaching

Undergraduate Teaching Assistant Jan 2024 – Dec 2024
University of Florida CISE Department *Gainesville, FL*

- Assisted with COP 4020: Programming Language Concepts, a course on programming language design and implementation.
- Held five hours of office hours per week, assisting students with programming assignments and conceptual understanding.
- Proctored quizzes, graded assignments, and monitored potential academic dishonesty in student repositories.

Open Source Contributions

Core Maintainer

- **Manhattan Reasoning Gym** – FPGA synthesis and agent-sandboxing toolkit, published on PyPI. pypi.org/project/manhattan-reasoning-gym
- **Wyvern Graph** – LLVM IR to knowledge graph translator. github.com/cscaff/Wyvern-Graph

Contributor

- **AlgebraicJulia** – Applied category theory in Julia; contributed to ACSets.jl, AlgebraicOptimization.jl, Catlab.jl, and DiagrammaticEquations.jl. github.com/AlgebraicJulia

Skills

Languages	Python, C/C++, Julia, Java, TypeScript, SQL (MariaDB)
Hardware Design	SystemVerilog, Amaranth HDL, SystemC, LiteX, VexRiscv
EDA & FPGA	Yosys, nextpnr, Siemens Catapult HLS, Intel Quartus, Xilinx Vivado, Lattice ECP5, DE1-SoC
Formal Methods	CBMC, Spot, SMT Solvers, Temporal Logics
Compilers	LLVM (IR analysis)
Machine Learning	PyTorch, Scikit-learn, Pandas, NumPy
Infrastructure	Docker, FastAPI, Redis, nginx, AWS, DigitalOcean, Tailscale, React, Git
Spoken	English, Portuguese